# United States Federal Spending in Georgia

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*Data Mining M2*

## I. Dataset Description

The data set used in this study comes from "USASPENDING.gov", a website that collects and holds records of awards given to each state since 2008 [1]. It breaks down awards into multiple categories, including contracts, contract IDVs, grants, loans, direct payments, and other award types (Figure 1). Each award has several pieces of data giving context to the award, including the recipient name, awarding agency, amount granted, etc.

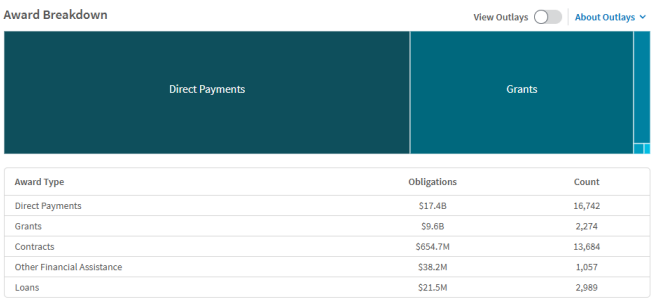


Fig. 1. A graph from "USASPENDING.gov" detailing the sum of all transaction types made to the state of Georgia from fiscal year 2008 to 2026.

This study specifically uses the web page that contains records of all federal awards granted to Georgia. This study only takes into account awards in 2025 totaling over 20,000,000 dollars, as this is the most relevant area for considering current spending. The primary attributes of each data point that will be examined are the recipient name, the awarding agency, the award type, the award subtype, the award description and the obligated amount (Table 1). The obligated value of the award was chosen over the outlay amount due to a significant amount of missing outlays present in the dataset.

While the obligated amount may reduce clarity on where federal money is going, it will still provide context into the interests of the federal government and their spending habits. There may also be a quality issue in the description of each award, as some are more clear than others, while some barely provide a description at all. For this reason, the description is not as considered for analysis as the rest of the data, however it is still provided to give context to each award. After sectioning out the most important portion of time, and carefully considering which pieces of data are most relevant to the study, the data set used contains 1,478 rows of entries, each with 6 columns of data.

| transactions | |
|---|---|
| transaction_id | string |
| recipient | string |
| awarding_agency | string |
| award_type | string |
| award_subtype | string |
| description | string |
| obligations | float |

TABLE I
Data schema describing the breakdown of each transaction and its relevant data. This study primarily deals in string values, with the exception of obligations, which are recorded as a float value.

## II. Discovery Questions

This data mining project aims to answer the following questions about U.S. federal spending:

- What combinations of awarding agencies, recipients, and award types frequently occur in different sized regions within Georgia?
- "What natural groupings can be found among federal funding recipients based on the characteristics of their transactions?"
- What characteristics distinguish anomalous high-value transactions from normal federal spending in Georgia?

In a time where U.S. citizens desire to know more than ever where their tax dollars are being used, answering these questions is an important task. It is vital to provide clarity and rationale for the average tax payer that their money is going to secure areas that benefit and strengthen their community. Unfortunately in the past few years, waves of questionable and fraudulent programs benefiting off of government subsidy have been brought to light, causing restlessness and conflict in America. Examining the state of Georgia should either provide reassurance to its citizens that their taxes are still primarily going to good places, or will expose an underlying problem within the state in need of correction.

## III. Planned Techniques

To begin the process, each transaction made within the relevant time period will be recorded within a data matrix

created using Python code (Figure 2). Once the data has been

```
transactionMatrix = [
    ["FA862506C6456","Lockheed Martin Corp","Department of Defense","Contract","Definitive Contract", "7,261,932,653.67"],
    ["2505GASMAP","GA Department of Community Health","Department of Health and Human Services","Grant","Block Grant", "12,849,962,570.00"],
    ["SLFRP1029","State of GA","Department of the Treasury","Direct Payment","Direct Payment for Specified Use", "4,853,511,033.91"],
    .
    .
    .
    .
    .
    .
    ]
    ]
```

Fig. 2. Sample data matrix demonstrating what the matrix will look like when it is recorded within the python script. This example demonstrates a recorded contract, a grant, and a direct payment.

consolidated into a table, two primary data-mining techniques will be used to analyze it. First, to discover what combinations of transaction characteristics occur in different regions in Georgia, the association rule will be implemented using Apriori. The data will be downloaded from the U.S. Spending website, and preprocessed into the data matrix. During this process, a new category will be added for ”county size”. Each county found in the data set will have its population recorded, as listed on the 2020 U.S. Population Census, which is the most official updated population record. After every county’s exact population is recorded, another new category will be created to record the population range. For this, every recorded population will be divided into equal sized categories, which are created based on the spread of different population numbers present. Once that category has been recorded, the apriori method will be applied to the data matrix, specifically examining the categories of awarding agency, award type, recipient, county name, and county size range. Once this is complete, the most frequent combinations will be examined to find different patterns in different counties and county size ranges.

After examining the rules generated from the apriori method, and determining the strength of the relations, the outliers discovered in the preprocessing step will be examined in order to determiner what characteristics define them over other transactions. For this, anomaly detection will be employed to further prune any values that are significantly higher or lower than the rest of the set. To do this, the IQR method will be used in context of the total obligations directed to the state of Georgia to confirm the flagging of these transactions as anomalies. Once we confirm the anomalies, we will use an isolation forest, where we will randomly select various features of each transaction, with the goal of determining which features cause the anomalies to happen.

## IV. Timeline

After the proposal is completed, the planned timeline will take into account three major milestone dates: M2 on March 5, M3 on April 2, and M4 on May 3. For M2, the goal is to completely implement the first data mining technique, which being anomaly detection, and to begin to implement the isolation forest. To accomplish this, preprocessing and data transformation will need to be complete before the data can be mined. The first half of M2 will be focused on accomplishing these tasks, along with setting up the data matrix within Python. At the same time, the preprocessing of the data and scripting of the apriori method will begin. The goal is to complete the initial setup of the data in the first two weeks, then complete and record the results of the apriori method in the second two weeks before March 5. After M2 is completed, the next major goal for M3 will be to implement the isolation forest and use it on the data matrix. The plan is to dedicate the first half of the milestone to scripting and debugging and to use the second half to run the data matrix through it and have a visualization of the resulting sets of data ready for interpretation. The last milestone, M4, will be dedicated to finalizing the interpretation and compiling it in the final report, ready to be presented to the wide audience. There will be 2 anticipated challenges throughout this process. The first is the time it will take to transfer the data into a data matrix for M2. There are over 1400 entries, which will be time consuming to accomplish. Despite this, by doing this in tandem with scripting the apriori method, and by pacing the creation in the span of two weeks, the matrix creation will be ready in time to be used. Secondly, with the sheer volume of transactions and recipients, it may be difficult to generate meaningful sets of common groups of characteristics. There is plenty of room for error here, since it takes more effort and analysis to match strings over integers, and the breadth of data makes this margin of error wider. However, given plenty of time to implement and debug the scripting, the accuracy of the analysis by the program should be close to guarantee.

## V. M2

Milestone 2 aims to answer the question: ”What combinations of awarding agencies, recipients, and funding types frequently co-occur in different regions of Georgia?” It examines records of every federal transaction made by the Federal government to Georgia, with a threshold of 20,000,000 dollars, within fiscal year 2025. The goal is to perform apriori to find out what different regions of Georgia (divided by population range) tend to receive the most in federal spending.

## VI. Exploratory Data Analysis and Preprocessing

When downloading the data from USASpending.gov, two sets of tables were received, one for contracts and one for grants and direct payments. Since contracts and grants work differently from each other, each table records different types of data. Despite this, both data sets contain the information we need to look for frequent combinations of awarding agencies and recipients in different regions, those being the awarding agency, recipient name, recipient county, and the transaction type. Even though these points of data are under different names, they can be combined under the same name.

We further checked the quality of the data, specifically looking for any missing values or duplicate transaction IDs. There appears to be some missing data, specifically several IDs that are null values. All of these missing ID values are linked to transactions listed as ”Multiple Recipients”. These are transactions, such as Social Security, that are delivered to a collective of individuals whose information is protected

by law, meaning these transactions are not actual null values. When the data is transformed for mining, these values will be replaced with filler ID values to mark each individual transaction.

Next, we checked the number of unique counties present in the data to determine if there was enough variety in county sizes to warrant exploring them as size ratios. There appeared

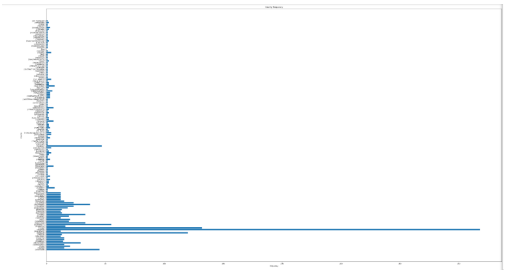

Fig. 3. Screenshot of the generated frequency graph for each county. There displayed 138 counties, however some were located outside of the state.

to be 138 counties, with the majority being smaller counties with a small number of transactions, while a hand full of larger counties received a larger number of transactions. This is perfect for determining what agencies have patterns of funding different sized counties. However, when exploring the counties even further, it was discovered that many of the counties listed were from different states. Many of these recipients have primary locations out of Georgia, but receive Georgia funding for Georgia locations. During preprocessing, we used the zip code listed on USASpending.gov to swap these counties out for their appropriate county. This left only seven transactions

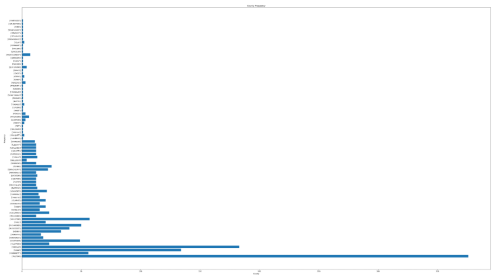

Fig. 4. Screenshot of the generated frequency graph for each county after the counties outside of Georgia were swapped.

without any listed Georgia zip code. With no way of retrieving the Georgia location for these seven recipients, they would only be noise in the algorithm, and were thus removed from the data.

Lastly, before preprocessing occurred, a box and whisker plot was generated to examine the number of outlier transaction values present in the set. The graph revealed that the top 25 percent of transactions were flagged as outlier values. The chances of the top fourth of the data being anomaly transactions was slim, so we needed to determine whether or not there were any anomaly transactions. To do this, a bar graph was generated that showed the total value

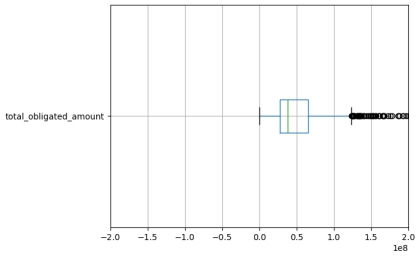


Fig. 5. Screenshot of the generated box and whisker plot for the total obligations.

of each transaction, from ascending order. This revealed a distribution with a gradual increase, and a spike only on the top four transactions, which represented transactions with high-support combinations. From this, it was decided that these transactions should not be removed from the table. There were three transactions marked with total obligations of zero. These are the only three transactions marked as ”loans”. Since loans are requested by the recipient and are paid back to the government, and these transactions would be naturally pruned by the algorithm, the three loans were removed from the data.

## VII. Transformation

After exploring the data, almost every necessary piece of data was present. The only column that needed to be added was the county population ranges of each transaction. To do this, another column was added, which recorded the exact size of each county, as listed in the most recent US Census from 2020. From here, we wanted to record the populations into equal sized bins, since we care more about common transaction attributes that relate to counties with similar attributes, rather than the counties themselves. After looking at the distribution of county populations, five bins were created: population 0 - 100,000, population 100,000 - 200,000, population 200,000 - 750,000, 750,000 - 950,000, and populations above 950,000. The spread of these bins are incredibly uneven, however this is due to the wide gap between population sizes that jump from 297595 straight to 764382. This makes the bin a similar size to the rest of the bins.

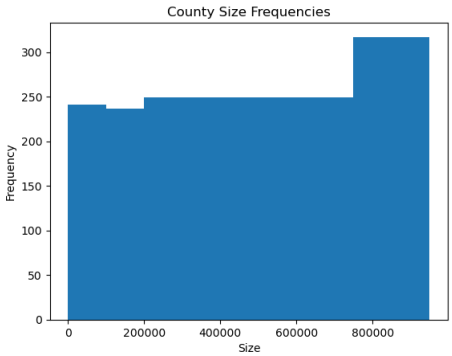


Fig. 6. Screenshot of the generated histogram displaying the frequency for every size ratio bin.

Once this piece of data was added, the final step before apriori is implemented is to list every transaction in a data matrix that can be run in apriori, containing only the relevant columns of data: awarding agency name, recipient name, recipient county name, county size ratio, and award type description. Each row of the matrix represents one transaction labeled with a unique ID from 0 - 1473.

## VIII. Data Mining

Once the data was combined into one excel file, and the county size and size ratios were added, they were placed into a 2D matrix, which would be the format used to generate the results of the apriori algorithm. Each row already represented a transaction, so the characteristics of each row were appended onto a list of strings, which would represent one transaction, then tat list was appended into the 2D matrix.

This matrix was then inserted into the apriori algorithm. We began by generating the frequent sets of data. A low support value of 0.01 was used due to the large number of recipients combined with the low number of awarding agencies. With a large number of transactions, even combinations that show up in 20 transactions will display significant information.

Within the collection of frequent sets of data, these are the main patterns that are being discovered:

- Direct payments from the Social Security Administration dominate the most common sets for every county ratio, excluding size 750,000 - 950,000. The administration appears commonly with size 0 - 100,000 with support value of 0.091, size 100,000 - 200,000 with support value of 0.098, size 200,000 - 750,000 with support value of 0.084, and size ¿950,000 with support value of 0.05. While these support values may appear small, they reveal large amounts of social security funding in all of these regions, ranging from 50 - 150 transactions each.
- Regions of size 750,000 - 950,000 appear the most in receiving delivery orders from the Department of Defense, with support of 0.039. This is due to the presence of Lockheed Martin Corp in Cobb County, which is a top recipient of contracts, and therefore skews the award type frequency in this area. The only other non-social security awards that frequently appear in this region are awards made to Emory University in DeKalb, with delivery order contracts appearing at a support value of 0.021 and cooperative agreements appearing at support value of 0.016.
- Only larger regions appear to receive different grant types from a larger variety of agencies. Of the sets with relevant support, Fulton is the only county to receive block grants (Department of Health and Human services with support value 0.023) or formula grants (Department of Education at support value of 0.016, Department of Transportation at support value 0.035, and Department of Agriculture at support value 0.011).
- The Department of Veterans Affairs appears commonly in all regions, however unlike social security, they appear more frequently in smaller regions (Size 100,000 - 200,000 at support value 0.022 and 200,000 - 750,000 at support value 0.029, specifically in regions closer to size 250,000).

The association rules were then generated to help support the common sets found initially. A confidence threshold of 0.8 was used to filter out any rules that would exist due to coincidence. The goal is to discover natural patterns of funding that stay consistent over time.

Hundreds of rules were found that helped support the validity of the frequent sets. Larger counties had a more variety of awarding agencies and funding types compared to more rural areas, with rules displaying confidence that ranged from 0.8 - 1, with lift values between 3 and 4. This is expected, as larger counties with large cities tend to house the majority of larger businesses and government organizations. Smaller counties had almost exclusive frequency with the Social Security Administration and the Department of Veterans Affairs. Each rule that displayed one of these two agencies given one of the smaller counties had a confidence value of 1, and a lift value averaging 6. This reveals the sheer amount of individuals in Georgia that receive government benefits related to retirement programs, and furthermore reveals the number of veterans in smaller areas that benefit from government support, which is a surprising metric compared to urban areas.

## IX. Interpretation

When examining all of the major findings from the frequent sets and rules, the majority of findings were trivial, and simply suggest common geographical patterns or common knowledge. The biggest pattern discovered was the large amount of social security transactions distributed in every region of Georgia. This suggests that there are a high amount of retired individuals and families within the state that receive social security or medicaid. The large amount of veterans affairs transactions in smaller counties might suggest a large amount of veterans who receive benefits from multiple government programs. However, without comparing these results to another state, such as Alabama, there is no way of knowing if these numbers are especially high compared to other areas in the country, and without the public ability to see every individual who collects these benefits, there is no way of determining how many of these transactions are legitimate and how many are fraudulent. This discovery does not reveal any differences among Georgia regions in regards to funding patterns.

The large variety of transactions in large regions, such as Fulton County and DeKalb county, is a relevant pattern, however it simply reaffirms that larger businesses and government organizations are found in more urban areas with lots of infrastructure and high population. It is no surprise that they would skew the support values for sets regarding awarding agencies and county sizes. Another factor that appears to skew the support for frequent sets is the existence of military-based operations, such as Lockheed and those found in Warner Robins. Both exist in different sized regions, however they receive a lot of Georgia's federal funding through various contracts. This does not help determine the difference between

different sized regions, rather it reaffirms the existence of important industries, such as military-based businesses, in larger regions.

## X. Limitations

While the rules generated by the apriori algorithm are relevant with strong support, the sheer volume of transactions and different recipients makes it difficult to find patterns that are not obvious. The descriptions of each transaction vary heavily, making it nearly impossible to use them as a factor in finding frequent sets of transaction characteristics among different regions. Furthermore, without examining other states in the same manner, it is difficult to display the weight of the significant rules found here. In the future, this same algorithm should be performed in other states of similar size in order to compare their sets. This would better reveal of the high number of social security transactions is exclusive to Georgia, or if they are the largest distribution of federal funds in the country.

Another limitation that may have effected the results is the categories used to create the sets. The county names were kept in addition to their sizes in order to help explain common sets further. However, with the imbalance in counties per size ratio, some sets appear as a consequence of county characteristics being paired with their names. Furthermore, certain recipients only receive specific types of funds from specific agencies. While the less frequent recipients were naturally filtered out, this lead to frequent sets appearing with identical information, but different support values. This was mitigated by favoring certain category combinations over others, such as awarding agencies paired with county sizes, however it is possible that this could have skewed the results and led to the large amount of trivial patterns.

## XI. Updated plans for M3

The original plan was to employ anomaly detection using an isolation forest to explain the causes of abnormally large transaction values. However, when exploring outliers in this milestone, it revealed that the largest transactions did not have abnormal characteristics when compared to other transactions. Furthermore, I want to shift into a different approach that may reveal more surprising information than the frequency sets generated by the apriori method. I am shifting my focus onto comparing the characteristics of the recipients themselves as opposed to the geographic regions they appear in. To do this, I will be using K-Means to cluster recipients based on the total amount of funds they received in 2025, the amount of funds dedicated to disaster relief, the amount of awards they received, and the amount of agencies that fund them. Performing this method will help answer the question, ”What natural groupings can be found among federal funding recipients based on the characteristics of their transactions?”

## References

[1] ”State Profile: Georgia.” *USAspending.gov*, https://www.usaspending.gov/state/Georgia. Accessed January 15,2026.